Team Project: Assessing the Impacts of Monetary Shocks in Canada and the USA

Elijah Adrian, Sophia Scarcella, Matthew Kielar, Amitaditya Paul and Florent Samson

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Data and Sample

We estimate separate structural VAR models for Canada and the United States using monthly data. Our baseline estimation sample spans 1985M1 to 2007M1 for the U.S. estimation and 1997M1 to 2020M1 for the Canada estimation. For the U.S. model, we choose to begin the sample in 1985 for data availability purposes. We close the sample in 2007 to avoid the Global Financial Crisis, to remove short-term volatility, and to avoid zero lower bound issues. The Canadian sample starts in 1997 for data availability reasons, and ends in 2020 to avoid the COVID period which created exceptional circumstances. We include the GFC to increase our sample size but use a dummy variable to control for its effects. U.S. macroeconomic series are obtained from the Federal Reserve Bank of St. Louis (FRED), and Canadian data are sourced from Statistics Canada, the Bank of Canada, and FRED. Oil price and exchange rate data are obtained from FRED as well. Data on excess bond premiums are sourced from the Federal Reserve Board. Where available, data series are seasonally adjusted.

For each country, the VAR includes measures of real activity, prices, labour market conditions, and the policy rate for the baseline version. We extend the model by including additional macroeconomic and financial variables. The extended model for Canada includes oil prices and exchange rates. The extended model for the U.S. includes oil prices, excess bond premiums, and inflation expectations. For robustness, we add a dummy variable for the Global Financial Crisis (GFC) in the Canadian model, and we use a high-frequency instrument in a proxy-VAR for the U.S. model.

For Canada, we use Gross Domestic Product (GDP) at basic prices in chained 2017 dollars for real activity, the 12 months growth of Consumer Price Index median (CPI-median) for prices, unemployment rate from the Labour Force Survey (LFS) for labour market conditions, and the Bank of Canada overnight rate annualized for policy rate. For oil prices, we use spot prices of West Texas Intermediate (WTI) crude measured in dollars per barrel. For exchange rates, we use spot prices of the CAD/USD currency pair. For robustness, we create a dummy variable that takes a value of 1 during the Global Financial Crisis (GFC), defined as the period between 2007M1 and 2009M12, and 0 otherwise. Prior to estimation, we perform some preliminary modifications to our data to ensure that stationarity is appropriately accounted for. We use the level for the overnight rate and inflation rate, first differentiation for the exchange rate, unemployment rate, and oil prices, then we use log first-differentiation for GDP.

For the United States, we use the industrial production index (base 2017) as a proxy for real activity, core CPI, defined as the CPI for urban consumers excluding food and energy, as a measure of prices, the unemployment rate for labour market conditions, and the federal funds rate annualized for policy rate. Oil prices are WTI crude spot rates. The measure of inflation expectations utilises the median expected price change in the next 12 months, obtained from the University of Michigan's Survey of Consumers as a measure of inflation expectations. For excess bond premiums, we use measures from the Board of Governors of the Federal Reserve. For robustness, we also use a high-frequency instrument for the policy rate as proposed by Gertler & Karadi (2015). To ensure stationarity, we use levels for the unemployment rate, federal funds rate, expected inflation, and excess bond premium. We use log first-differentiation for industrial production, core CPI and oil prices.

Identification

Baseline Model

We identify monetary shocks using a recursive Cholesky decomposition, which imposes short-run impact restrictions through a chosen ordering of variables. In the baseline model, we order the following variables: log real GDP, CPI (median), unemployment rate, and the overnight rate. Under this approach, variables earlier in the ordering are assumed not to respond contemporaneously to shocks affecting variables later in the ordering. In contrast, later variables respond immediately (to themselves and) to earlier variables. Real activity (log real GDP) is ordered first and is therefore assumed not to respond contemporaneously to shocks to any other variable in the system (inflation, unemployment, and the policy rate). The effect on measured inflation typically comes with a lag, reflecting price stickiness and the fact that inflation measures aggregate many individual price changes. We therefore place log real GDP before median CPI, allowing output innovations to affect inflation contemporaneously, while ruling out an immediate within-period effect of inflation innovations on real output. Unemployment is ordered next as a relatively slower labour market outcome. Firms typically respond to shocks by changing hours and their utilization of employed labour (after deciding on output), which affects the unemployment rate. This occurs more gradually due to hiring/firing costs and matching frictions. Finally, the overnight rate is ordered last to allow monetary policy to respond contemporaneously to innovations in real output, inflation, and labour market conditions, which is consistent with the central bank monitoring these indicators when setting the overnight rate.

Extended Model

For Canada, the extended model makes two additions to the baseline: the CAD/USD exchange rate as an endogenous variable and oil prices as an exogenous variable. Oil prices are treated as exogenous rather than ordered within the Cholesky system because Canada is a small open economy that is a price-taker in global energy markets. WTI prices are determined by global supply and demand dynamics that Canadian macroeconomic conditions have no meaningful influence over. Treating WTI as exogenous is therefore more logical than placing it first in a Cholesky ordering, which would still allow feedback from Canadian variables to oil prices at longer horizons. The CAD/USD exchange rate is included as an endogenous variable because it is a critical transmission channel for both external shocks and monetary policy in an open economy. This is because oil price movements affect the CAD through Canada's petrocurrency relationship (higher oil prices imply greater demand for CAD, resulting in a currency appreciation), while changes in the overnight rate affect the exchange rate within the period through interest rate differentials and capital flows under uncovered interest parity. Omitting the exchange rate from the baseline model leaves this transmission channel unaccounted for and can produce distorted impulse responses.

For the U.S. model, we add 12 months-ahead expected inflation and excess bond premium as endogenous variables. We rank them right before federal funds. Inflation expectations are endogenous and ranked highly in our ordering because they react quickly to commodity prices, current inflation, and other macroeconomic news. Long term inflation expectations are usually anchored at the central bank's target when credibility is high, but short-term expectations are highly reactive and correlate with the business cycle and policy decisions. Information about inflation, oil, real activity, or unemployment all affect expectations immediately. Similarly, the excess bond premium is a financial variable containing the excess risk on bond premiums once

individual risk is removed, leaving only structural risks. This structural risk depends on all other macroeconomic variables and responds quickly, as it is a financial variable. However, neither of these variables affects the real economy contemporaneously, so they are ranked after measures of real activity, oil, inflation, and the unemployment rate.

Results

Canada

The baseline model IRF shows the response of the Canadian economy to a positive shock to the overnight rate over a 36-month horizon with a 12-month lag specification. The overnight rate itself behaves as expected, jumping on impact to roughly 0.13 percentage points (pp), peaking around 0.23 pp by months 3 to 4, and then decaying slowly but remaining statistically significant through approximately month 25 before fading toward zero, reflecting the gradual and persistent nature of Bank of Canada tightening cycles. The most problematic result is the CPI median, which exhibits a clear price puzzle as first described by Sims (1992): rather than falling in response to monetary tightening, inflation rises immediately, peaks at around 0.04 pp near months 8 to 10, and remains positive and statistically significant for nearly the entire 36-month horizon. Sims (1992) attributed this phenomenon to omitted variables that the central bank monitors when setting policy, and Ha et al. (2025) more recently formalize this as a forward-looking expectations problem, where standard VAR models fail to capture future inflation developments, so the Bank of Canada's anticipatory policy responses are misidentified as exogenous tightening shocks, producing a spurious positive correlation between the overnight rate and inflation. The log real GDP response is statistically insignificant throughout, with a small positive jump on impact that quickly turns slightly negative around months 2 to 4, then oscillates around zero with wide confidence bands. This fails to produce the hump-shaped negative response documented by Christiano, Eichenbaum, and Evans (1999) as the benchmark for a well-identified monetary policy shock. Similarly, the unemployment rate briefly dips negative on impact before returning to zero, which is counterintuitive since tightening should raise unemployment as economic activity slows, though the response is statistically insignificant throughout.

The extended model IRF, which adds WTI oil prices as an exogenous variable and the CAD/USD exchange rate as an endogenous variable, produces broadly similar results across the 36-month horizon. The overnight rate response is (mostly) unchanged from the baseline, peaking around 0.21 to 0.23 pp by months 3 to 4 and decaying gradually, which reassures us that the policy shock itself is robustly identified across both specifications. However, the price puzzle in CPI median persists as inflation still rises on impact, peaks near 0.04 around months 8 to 10, and remains positive and statistically significant for most of the horizon. This is notable because Cushman and Zha (1997) demonstrated that explicitly accounting for Canada's open economy structure resolves the price puzzle in Canadian VARs, and Kim and Lim (2018) similarly show that the price puzzle disappears in small open economies, including Canada once the foreign exchange (FX) puzzle is resolved. The persistence of the price puzzle here is therefore likely connected to the CAD/USD response, which is statistically insignificant throughout, with the point estimate oscillating around zero. Hence, the model does not find a clean appreciation of the CAD following a rate hike as UIP would predict, implying the disinflationary exchange rate channel is not operating, which Ha et al. (2025) identify as a key mechanism linking the FX puzzle and the price puzzle. The log real GDP response remains insignificant and broadly mirrors the baseline, again failing to produce the hump-shaped contractionary response of Christiano, Eichenbaum, and Evans (1999). The unemployment

rate again goes negative on impact before recovering, also remaining statistically insignificant from zero throughout. Taken together, the persistence of the price puzzle across both models suggests that, consistent with Ha et al. (2025), the omission of forward-looking inflation expectations may be the deeper source of misidentification that neither the baseline nor the extended model fully addresses.

United States

The baseline model for the US specification shows the impact of a one standard deviation contractionary shock to the federal funds rate on industrial production, unemployment rate, and core CPI over a 36-month horizon with a 6 lag specification. We found the 6 lag specification produced a smoother and less volatile output than the 12 lag specification in the US context. The federal funds rate is shocked by around 0.17 pp, and increases up to 0.27 by month 7, then follows a constant decrease for the rest of the projection period. It stays significantly positive until month 32, where the confidence interval crosses 0. Industrial production, as specified in our identification assumptions, is not affected contemporarily by other variables and thus begins at zero. It first spikes to 0.08 pp in the first month and is significantly positive; the impact quickly returns to zero, showing no other significant response to the shock over the whole projection period. Similar to the Canadian specification, the response of core CPI is emblematic of the price puzzle. The impulse response is very volatile in the first 10 months, but settles significantly positive by about 0.05 pp from months 11 to 25, after which it becomes insignificant. Here again, omitted variables and a misidentification of forward-looking expectations from the Fed possibly create a spurious positive correlation between the policy rate and inflation. The unemployment rate also fails to reproduce the expected results. From theory, we expect the unemployment rate to rise after a monetary policy shock. In our SVAR estimation, the unemployment rate declines for the first 10 months, reaching around -0.05 without being significantly different from zero. Subsequently, the unemployment rate response rises, crossing the zero line after month 30, still without being significant. We believe the same reasons behind the price puzzle are acting on the unemployment rate impulse response function. Inflation is often associated with a low unemployment rate, and thus, preemptive actions to restrain inflation not being accounted for would also cause a negative shock to unemployment.

We extend the model to mitigate some of the issues behind this price puzzle. We add WTI oil prices as a proxy for commodities' price dynamics, the excess bond premium to capture financial market conditions, and inflation expectations to include some of the anticipatory policy responses. In this extended model, the impulse response function of the fed fund rate is similar to the baseline since it is the one being shocked in our models. Industrial production shows a muted response, spiking only to 0.06pp before returning to equilibrium. Adding new variables helped reduce the amplitude of our price puzzle, but did not transform the core CPI and unemployment rate into their theoretical impulse response functions. Core CPI follows the same volatility but settles insignificantly at 0.005 pp, while the unemployment rate's response now crosses into positive territory around month 22 but remains insignificant. The excess bond premium shows a positive response at around 0.04 pp, but it isn't significant beyond a few months between months 20 and 30. Corresponding to our identification, oil prices appear exogenous, as they show no response beyond initial volatility. Lastly, inflation expectations show an initial positive response of around 0.04pp but then return to equilibrium quickly. Sims (1992) found that a commodity price index acted as a forward-looking variable, capturing early signs of price pressure that central banks were reacting to, and that including it in SVAR estimations mitigated the price puzzle. We thought that oil prices would be a good measure for commodity prices and thus would help solve the omitted

variable bias behind the price puzzle. Similarly, our theory was that using inflation expectations should capture some of that forward-looking information. However, these new variables did not reverse the price puzzle but did reduce the amplitude of their response. We think that part of why all impulse response functions from the extended model are more muted versions of the baseline model is that the new variables partly reduce the price puzzle, but also introduce more degrees of freedom, absorb part of the shock, and may be too correlated with other variables. Core CPI inflation and inflation expectations are largely collinear, and in our model, it is hard to specify the ordering of those two variables, since expectations are shaped by today's level, and the level is a function of expectations, as we have seen in the neoclassical model. We conduct several robustness checks in the following section.

Robustness Checks

Canada

To assess the robustness of the extended model, the Global Financial Crisis (GFC) is introduced as a dummy control variable and time interaction capturing the period from 2007 to 2009. The GFC represented a considerable disruption to credit markets, financial conditions, and monetary policy transmission that falls well outside the normal variation that the VAR is designed to capture. Not accounting for it risks conflating the structural effects of the crisis with the estimated monetary policy shock. When comparing the GFC-controlled IRF to the extended model, the results are remarkably stable: the overnight rate follows a near-identical path, peaking around 0.20 to 0.21 pp by months 3 to 4 and decaying gradually. The price puzzle in CPI median persists, though the point estimate is about 40% lower and the lower confidence band crosses zero earlier around months 20 to 22, suggesting the GFC control provides a slight but meaningful improvement in the inflation response. The CAD/USD, log real GDP, and unemployment responses remain statistically insignificant throughout, broadly mirroring the extended model without the control. The consistency of results across both specifications suggests the core findings are not driven by the GFC period and that the extended model is reasonably robust to this major macroeconomic disruption.

United States

We conduct several robustness checks to ensure the stability of our results. First is using 12 lags in identification instead of 6. We found that it led to the same results over the middle and later parts of the 36-month estimation, but produced more volatility in the first few months. Using the all-items CPI instead of the core CPI resulted in larger and more positive responses for inflation, but didn't materially change the direction of any impulse response function.

In addition to the recursive short-term restrictions identification method, we used an instrumental variable method using a high-frequency measure. This instrument is made of small movements in interest rate futures in a tight schedule around FOMC announcements. These surprises are then regressed on the macroeconomic information available prior to a rate decision, and the residuals become our IV. This instrument is assumed to plausibly isolate monetary policy shocks and is untainted by other macroeconomic developments (Gertler and Karadi 2015). Using this instrumental variable, we run a proxy VAR on the extended model. This new identification method gives us similar results for industrial production, core CPI, oil prices, and excess bond premium. However, the impulse response of inflation expectations becomes negative in the 14th month, and

the unemployment rate's impulse response becomes clearly positive around month 17th, after an initial decline. While this new identification method doesn't solve the price puzzle, it does help shape the impulse response functions closer to theory.

Conclusion

We estimate two different structural VAR, one using Canadian data and one using American data, to estimate the effects on monetary policy shocks. In each case, we estimate a baseline model and an extended model using additional macroeconomic variables. We observe different results from the existing literature and neoclassical theory. Real activity becomes volatile in the short term, and shows no response over the medium and long term. Prices and unemployment move in the opposite direction from what we expected, with prices rising and unemployment declining following a contractionary monetary policy shock. This is emblematic of the price puzzle, identified in previous literature. Adding known fixes to this puzzle such as commodity prices and forward looking inflation expectations only mitigated the amplitude but did not reverse our results. For robustness, we tried different specifications including a proxy VAR using a high frequency measure of monetary policy, but the results stayed broadly similar to our baseline.

References

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Appendix

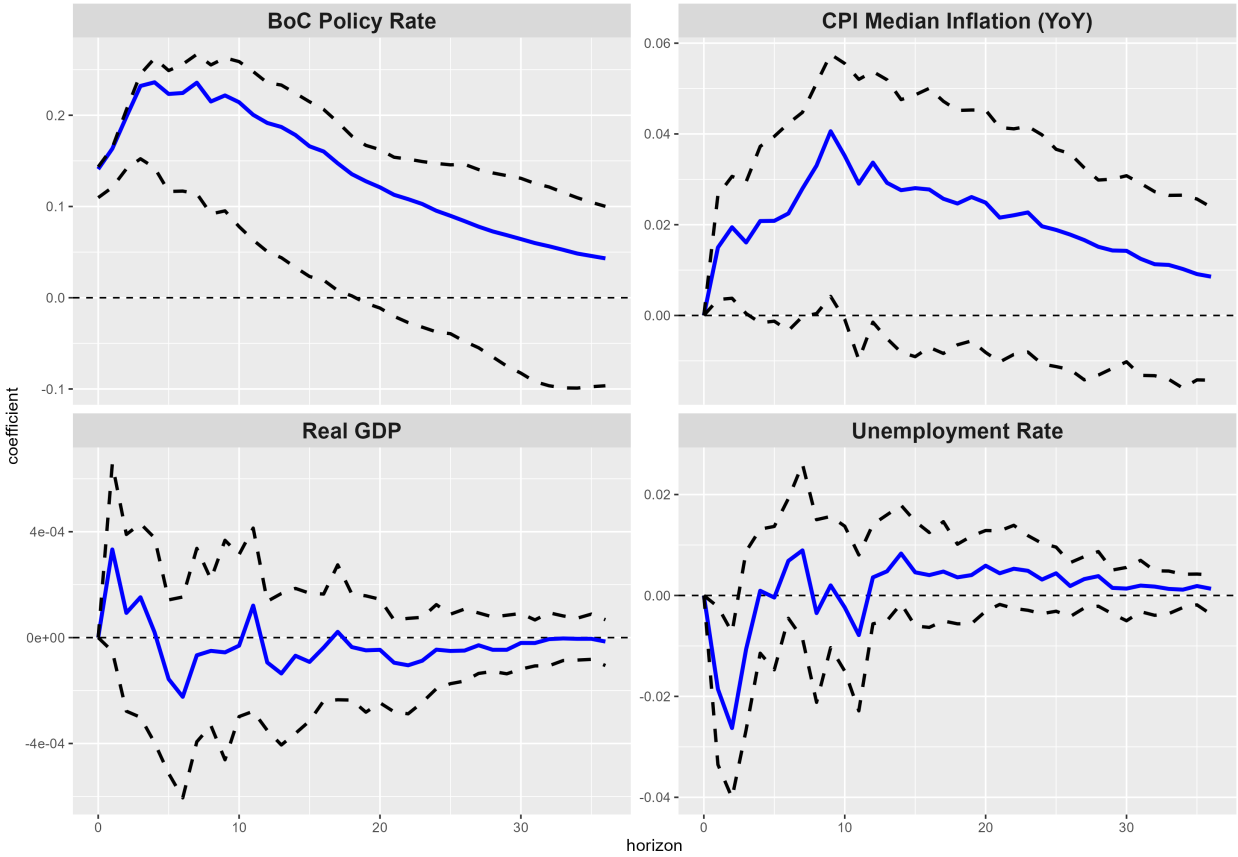

Figure 1: Orthogonalised Impulse Responses of a Monetary Shock in Canada (Baseline, 12 lags)

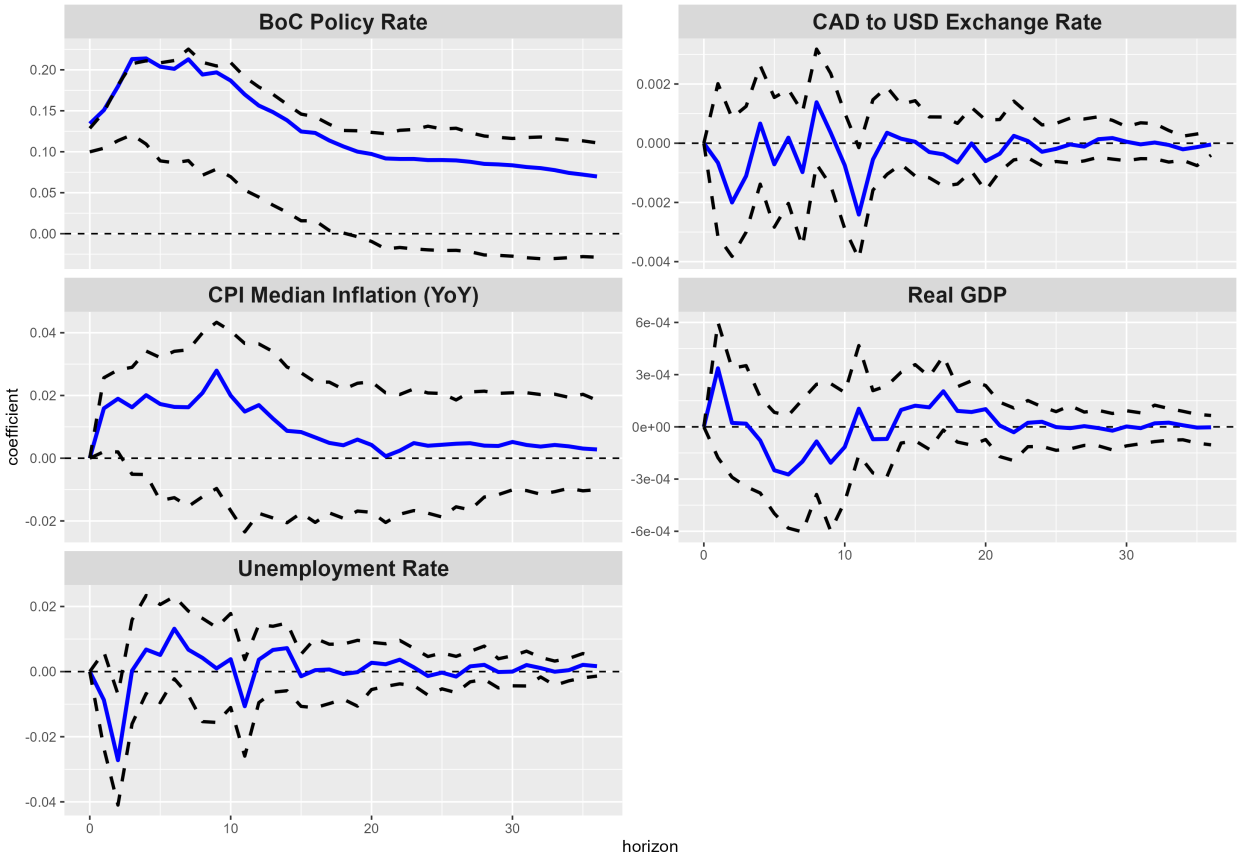

Figure 2: Orthogonalised Impulse Responses of a Monetary Shock in Canada (Extended, 12 lags)

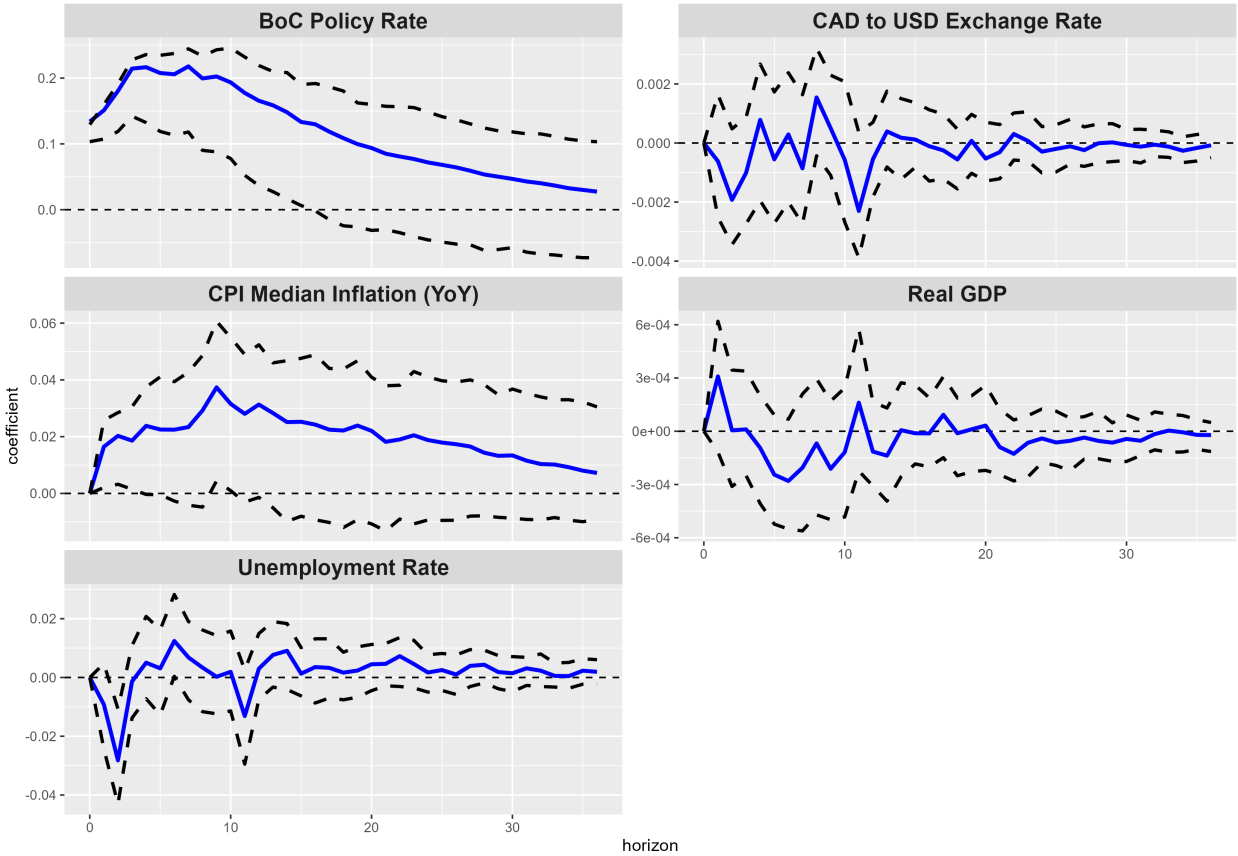

Figure 3: Orthogonalised Impulse Responses of a Monetary Shock in Canada (Extended, with GFC controls, 12 lags)

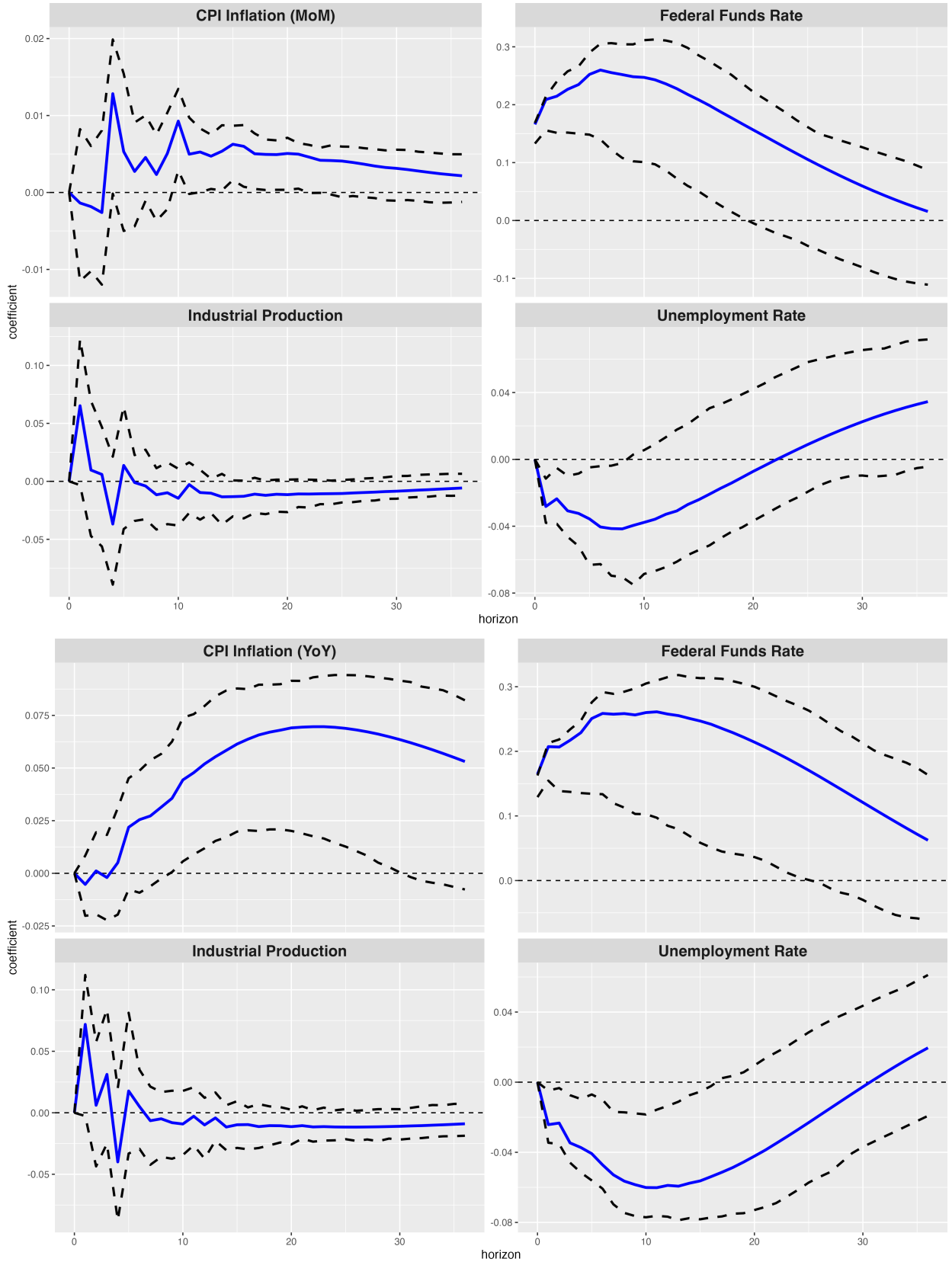

Figure 4: Orthogonalised Impulse Responses of a Monetary Shock in the U.S. (Top: MoM CPI, 6 lags; Bottom: YoY CPI, 6 lags)

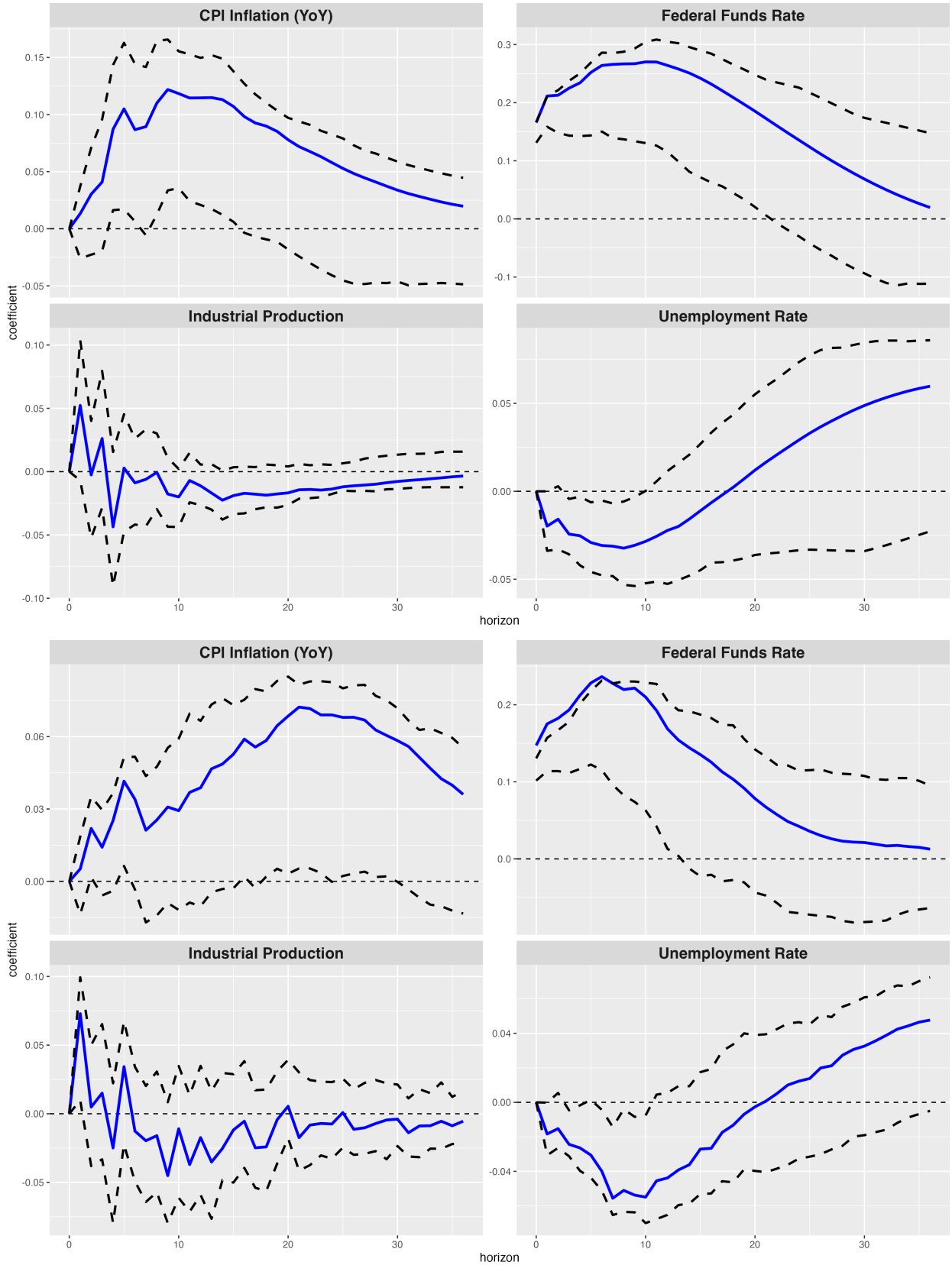

Figure 5: Orthogonalised Impulse Responses of a Monetary Shock in the U.S. (Top: Headline YoY CPI, 6 lags; Bottom: YoY CPI, 12 lags)

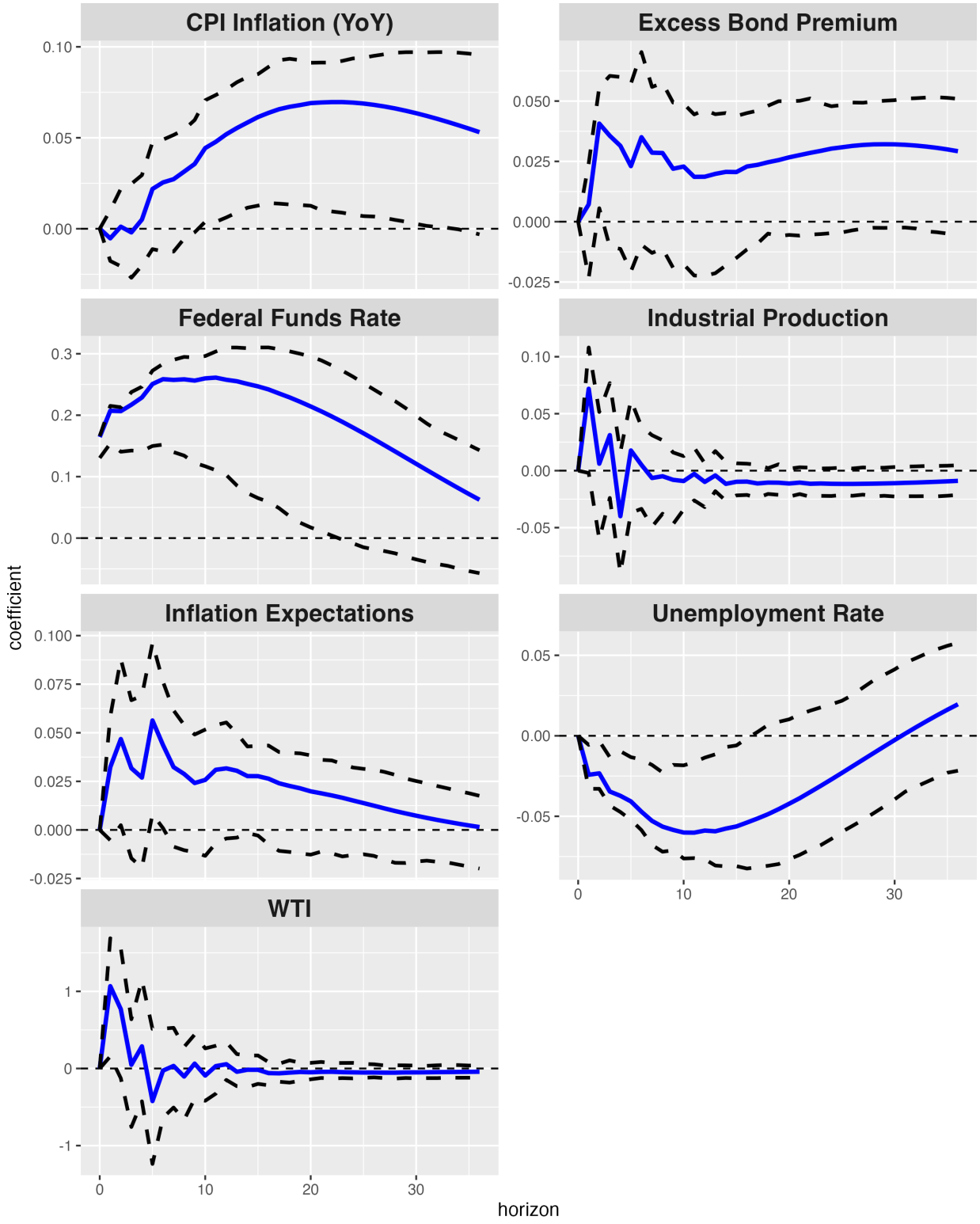

Figure 6: Orthogonalised Impulse Responses of a Monetary Shock in the U.S. (Extended Model)

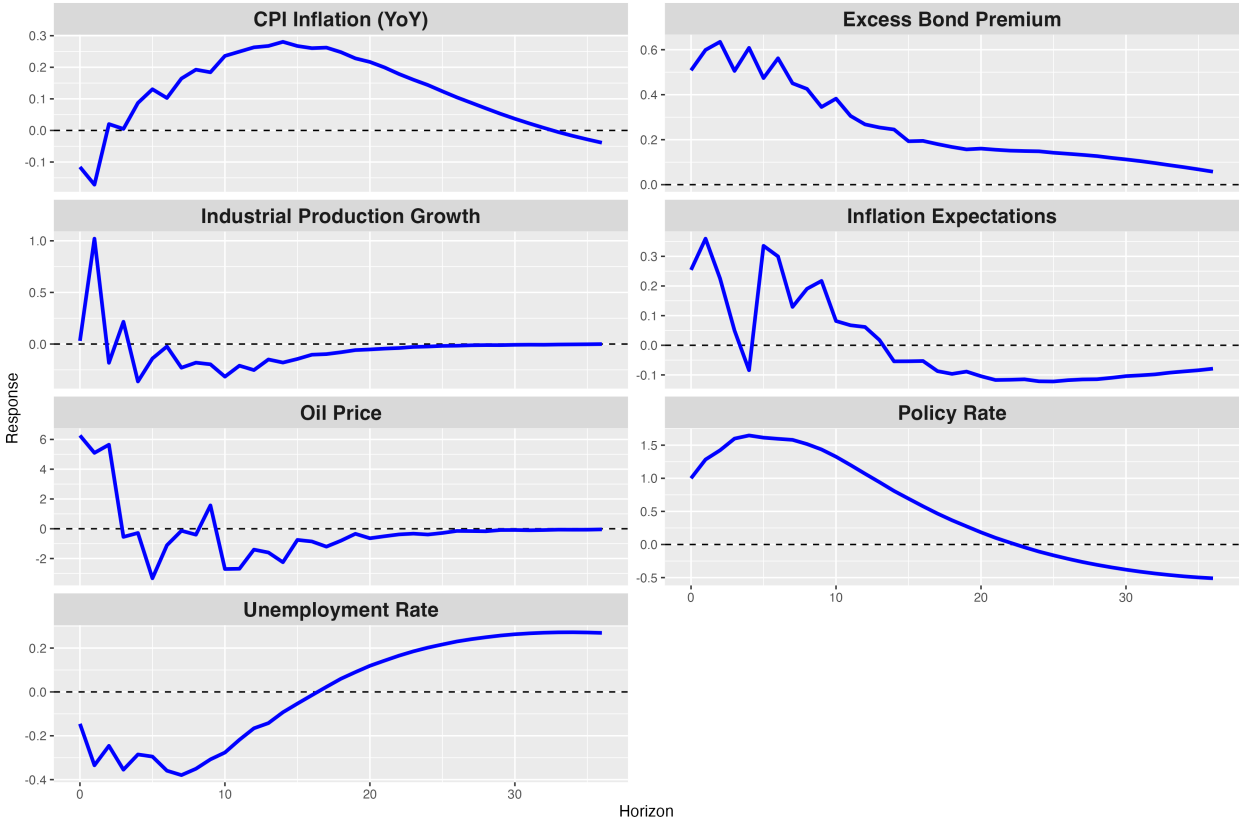

Figure 7: Impulse Responses of a Monetary Shock in the U.S. (High-Frequency Identification)